

Griffin Burke

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EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

Master of Computer Science | GPA: 4.0/4.0

Jan 2027 – Dec 2027

Bachelor of Science in Computer Science, Minor in Mathematics | GPA: 4.0/4.0

Aug 2023 – Dec 2026

Coursework: Artificial Intelligence, Deep Learning for Computer Vision, Database Systems, System Programming, Algorithms and Models of Computation, Data Structures, Distributed Systems (FA26), ML Systems (FA26)

Course Assistant: Numerical Methods (FA26), Probability and Statistics (FA25, SP26), Linear Algebra (SP25, SU25)

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, C, C++, Kotlin, Java, HTML/CSS

ML & Data: Azure Databricks, LangChain, PyTorch, Scikit-learn, pandas, NumPy, Keras, HuggingFace, Ultralytics

Software & Cloud: Flask, FastAPI, Express, React, Svelte, GCP, MySQL, Git

EXPERIENCE

Data Science Engineer Intern, AT&T

Jun 2026 – Aug 2026

- Engineered an orchestrator-subagent system to synthesize 15+ data sources, driving discovery of stale IP allocation
- Designed a data aggregation pipeline that builds agent context from a single source of truth, centralizing changes across category-specific subagents
- Integrated an existing API into a Databricks MCP server, enabling scoped agent search of IP address information

Undergraduate Research Assistant, Illinois Digital Animal Systems Lab

Jan 2025 – Present

- Developed a computer vision system for frame-level cattle weight estimation in a precision livestock platform
- Fine-tuned YOLOv11 on 800 segmentations of overhead cattle RGB-D images, achieving an mAP50-95 of 0.982
- Designed a frame conditioning pipeline for misaligned and noisy frames, reducing weight prediction RMSE by 4.5%
- Recovered 350,000+ training images by reconstructing a 200+ GB file containing corrupted sensor data

AI Team Lead, Kashmir World Foundation

May 2025 – May 2026

- Architected an edge ML system combining video, audio, and environmental data to detect poachers in the forest
- Directed development of ML models for audio filtering, bird-call classification, and poacher detection
- Partnered with hardware and simulation teams to define power and compute constraints for edge devices, jointly developing distributed ML pipelines and inter-device communication protocols

Machine Learning Engineer Intern, Kashmir World Foundation

Sep 2024 – May 2025

- Engineered a data pipeline to align and aggregate audio, weather, and experiment data for downstream modeling
- Achieved a base precision of 0.885 and recall of 0.882 using YOLOv8 for species detection on trail cam images
- Fine-tuned the model via hyperparameter sweeps and data augmentation, boosting precision 3.5% and recall 2.8%

PERSONAL PROJECTS

Note Transfer | Python, TypeScript, FastAPI, Svelte, TailwindCSS

May 2025 – Jan 2026

- Built an AI system using YOLOv11 vision and Microsoft TrOCR models to convert handwritten notes into markdown
- Engineered a multi-stage vision pipeline that detects, segments, and hierarchically organizes scanned notes into a standardized document tree

UIUC Involvement Bot | Python, Flask, Keras, HTML/CSS/JS, SQLite

Feb 2024 – Jun 2024

- Collaborated closely on the end-to-end development of a club recommendation website for UIUC students
- Developed, trained, and tested two neural network models with Keras, achieving accuracies of 89.4% and 95.6%